

Odonata Survey

Swadini, Blyde River Canyon

Mpumalanga · South Africa



Microgomphus nyassicus (Eastern Scissortail)

Seasonal observations 2019 – 2025

3,396

TOTAL RECORDS

70

SPECIES

7

SEASONS

9

FAMILIES

Anisoptera (Dragonflies)

46 species · 4 families

Zygoptera (Damselflies)

24 species · 5 families

Observer

Andries de Vries · Citizen Scientist · Middelburg, Mpumalanga · Observation.org

1. Introduction



Platycypha caligata — Dancing Jewel · Photo: Andries de Vries *Nesciothemis farinosa* — Eastern Blacktail · Photo: Andries de Vries

This report summarises seven consecutive seasons of dragonfly and damselfly (Odonata) observations conducted by citizen scientist Andries de Vries at **Swadini Forever Resort**, situated within the Blyde River Canyon Nature Reserve, Mpumalanga, South Africa. The study area covers approximately 1.1 km² on the western bank of the Blyde River, directly below the Blyderivierspoort Dam, within the canyon itself.

Odonata — the dragonflies and damselflies — are widely regarded as reliable bioindicators of freshwater ecosystem health. Their presence, diversity, and seasonal patterns reflect water quality, riparian vegetation condition, and the integrity of aquatic habitats. A rich and stable Odonata community is therefore a strong indicator of a well-functioning river system.

Observations were originally submitted to the Virtual Museum at the University of Cape Town (vmus.adu.org.za) and subsequently transferred to the international citizen science platform Observation.org. All records from 2019–2024 carry ‘Accepted (with evidence)’ validation status, confirmed by expert panels. New records from late 2024 and 2025 were submitted directly to Observation.org.

Table 1 — Survey Season Summary

Year	Records	Species	IOC Weeks	Notes
2019	580	58	18–31, 45–47	Two visits: May–Aug and Nov. Broad coverage including late season.
2020	287	46	33–42	Lockdown delayed arrival to August. Aug–Oct coverage only.
2021	682	55	16–35	Highest record count; most intensive survey season.
2022	442	49	17–34	Consistent effort throughout season.
2023	511	44	17–33	Good conditions. Strong early-season diversity.
2024	321	38	17–32	Below-average rainfall; noticeable decline in insect numbers.
2025	573	34	18–35	Data complete to 27 Aug. Mobile app enabled photo-free records.

Species counts are for identified species only (indet./spec. entries excluded). IOC Week numbers follow ISO 8601 (Monday-start). *Orthetrum julia falsum* (2025) normalised to *O. julia* throughout.

2. Species Diversity

A total of **70 identified species** across **9 families** were recorded over the seven seasons from 3,396 individual observation records. This represents approximately **43% of South Africa's national Odonata checklist** (~163 species) concentrated within a study area of only 1.1 km² — demonstrating exceptional biodiversity at a single canyon site.

The 70 species divide into two suborders: **46 Anisoptera** (dragonflies, across 4 families) and **24 Zygoptera** (damselflies, across 5 families). Libellulidae (skimmers, dropwings and allies) is the most species-rich family with 41 species, comprising 59% of the total. Among damselflies, Coenagrionidae (sprites, slims and allies) leads with 14 species.

Table 2 — Species per Family per Year

Suborder	Family	Group	2019	2020	2021	2022	2023	2024	2025	Total
	Aeshnidae	Emperors & Hawkers	3	3	3	2	2	1	2	3
Anisoptera	Gomphidae	Clubtails	5	1	4	4	5	3	3	7
	Macromiidae	Cruisers	1	1	1	1	1	1	1	1
	Libellulidae	Skimmers & Dropwings	29	24	22	22	18	17	15	41
	Calopterygidae	Demoiselles	1	1	1	1	1	1	1	1
	Chlorocyphidae	Jewels	1	1	1	1	1	1	1	1
Zygoptera	Lestidae	Spreadwings	1	2	3	2	2	2	2	3
	Coenagrionidae	Sprites & Slims	9	8	14	9	10	8	6	14
	Platycnemididae	Threadtails & Riverjacks	2	2	2	2	2	2	2	2

Per-year counts show identified species active that season. 2025 data complete to 27 August.

3. Core Resident Species

Twenty-four species were recorded in all seven consecutive seasons — including the shortened 2020 season (August–October only) and the drought-affected 2024 season. These are the **confirmed core resident fauna** of the study area, species reliably expected on any visit throughout the survey window.



Crocothemis erythraea — Broad Scarlet · Photo: Andries de Vries *Trithemis aconita* — Halfshade Dropwing · Photo: Andries de Vries

Table 3 — Core Resident Species (All 7 Seasons)

Suborder	Family	Group	Scientific Name	Common Name
	Aeshnidae	Emperors	<i>Anax speratus</i>	Orange Emperor
	Macromiidae	Cruisers	<i>Phyllomacromia picta</i>	Darting Cruiser
	Libellulidae	Skimmers	<i>Orthetrum chrysostigma</i>	Epaulet Skimmer
			<i>Orthetrum julia</i>	Julia Skimmer
		Blacktails	<i>Nesciothemis farinosa</i>	Eastern Blacktail
		Widows	<i>Palpopleura lucia</i>	Lucia Widow
Anisoptera		Elves	<i>Tetrathemis polleni</i>	Black-splashed Elf
		Dropwings	<i>Trithemis aconita</i>	Halfshade Dropwing
			<i>Trithemis arteriosa</i>	Red-veined Dropwing
			<i>Trithemis donaldsoni</i>	Denim Dropwing
			<i>Trithemis kirbyi</i>	Orange-winged Dropwing
			<i>Trithemis pluvialis</i>	Russet Dropwing
		Scarlets	<i>Crocothemis erythraea</i>	Broad Scarlet
			<i>Crocothemis sanguinolenta</i>	Little Scarlet

Suborder	Family	Group	Scientific Name	Common Name
	Calopterygidae	Demoiselles	Phaon iridipennis	Glistening Demoiselle
	Chlorocyphidae	Jewels	Platycypha caligata	Dancing Jewel
	Lestidae	Spreadwings	Lestes plagiatus	Highland Spreadwing
	Coenagrionidae	Sprites	Pseudagrion gamblesi	Great Sprite
Zygoptera			Pseudagrion hamoni	Swarthy Sprite
			Pseudagrion kersteni	Powder-faced Sprite
			Pseudagrion sjoestedti	Variable Sprite
			Pseudagrion sublacteum	Cherry-eye Sprite
	Platycnemididae	Threadtails	Elatoneura glauca	Common Threadtail
		Riverjacks	Mesocnemis singularis	Savanna Riverjack

Common names per Wilkinson & Dickinson, dragonflies-id.co.za. All 24 species recorded in all 7 seasons including 2020 (Aug–Oct only) and 2024 (drought year).

4. Seasonal Patterns & Phenology

In the Southern Hemisphere, the austral summer (January–April) represents the true Odonata peak — a period not covered by the current survey window, as the observer arrives in late April or early May. By the time surveys commence, species richness is already declining from its January–April maximum. The dataset captures the **late-summer decline, the winter trough, and early recovery**, but not the true annual peak.

Monthly pattern. Species richness peaks in **May (weeks 18–22)** with 35–42 species active. July (weeks 27–30) is consistently the lowest month — confirming the observer's field impression of mid-winter as the quietest period. August shows early recovery as temperatures rise.

Winter-active core. Despite the mid-winter reduction, 20–33 species remain on the wing through June–August. Twelve species were recorded in winter across all seven seasons — among them *Trithemis arteriosa*, *Orthetrum chrysostigma*, *Platycypha caligata*, and several *Pseudagrion* sprites. This year-round diversity makes Swadini an attractive destination for Odonata enthusiasts at any time during the survey period.

Under-sampled windows. The 2020 survey (Aug–Oct) and the 2019 November visit each produced 20–38 species per month, strongly suggesting September–November is as productive as May. Year-round surveys beginning in January would likely reveal substantially higher total species richness.



Trithemis arteriosa Red-veined Darning Green
Dropwing Photo: Andries de Vries

Table 4 — Monthly Species Richness by Year

Year	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Records
2019	—	35	25	22	13	—	—	38	580
2020	—	—	—	—	20	32	37	—	287
2021	34	42	33	26	25	—	—	—	682
2022	16	35	19	23	26	—	—	—	442
2023	27	34	26	24	17	—	—	—	511
2024	18	36	16	14	13	—	—	—	321
2025	—	29	22	16	16	—	—	—	573

Values are species richness per calendar month. — = month not surveyed. July values in red highlight the consistent winter low point.

5. Environmental Impact — The 2024 Drought



Lestinogomphus angustus — Spined Fairytail · Photo: Andries de Vries

Trithemis pluvialis — Russet Dropwing · Photo: Andries de Vries

The 2024 survey season stands apart from all others. Despite covering a comparable date range (weeks 17–32), it produced only **38 species and 321 records** — both the lowest of any high-effort season. For comparison, 2023 (same window, similar effort) yielded 44 species and 511 records.

Below-average rainfall during the late summer of 2024 and continuing into 2025 was accompanied by a conspicuous decline in birds, insects, and general invertebrate activity. Vegetation transitioned to its winter state earlier and more abruptly — signs consistent with drought stress in the riparian community. Every Odonata family showed reduced species richness in 2024.

The partial recovery in 2025 (34 species, 573 records by late August) is encouraging, though species richness remains below pre-drought levels. Several species absent in 2024 returned in 2025, suggesting condition-driven suppression rather than permanent local loss. Continued monitoring will be important to track the full recovery trajectory.

Note: Rainfall data for the study area has been meticulously recorded since January 2000 by Mr. Johan Pienaar, residing in the Blyderivier Botaniese Reservaat, less than 5 km from Swadini. Integration of this rainfall record with the Odonata dataset will be the subject of a future analysis.

6. Notable & Rare Species

Five species recorded at Swadini are considered nationally significant due to restricted range, rarity, or scientific importance. All are confirmed by photographic evidence.

Microgomphus nyassicus — Eastern Scissortail

Gomphidae (Clubtail) · Anisoptera · 43 mm · 54 mm wingspan

The most significant discovery of the survey. Previously known in South Africa only from exuviae (~1968, Oribi Gorge, KZN) and larvae (2018, Komati River, Mpumalanga). The first living adult ever photographed in South Africa was recorded at Swadini on 20 May 2019 by Antoinette Snyman (survey team), with the observer photographing another individual three days later at the same locality. Observed in the same area every season from 2021–2025, with copulating pairs (in copula) documented in 2021 and 2025 — confirming active breeding in the Blyde River Canyon.



Microgomphus nyassicus — mating pair (in copula), 2025 · Photo: Andries de Vries



Microgomphus nyassicus — male, dorsal view · Photo: Andries de Vries

***Pseudagrion gamblesi* — Great Sprite**

Coenagrionidae · Zygoptera · 45–50 mm

Nationally uncommon: confirmed by photographic evidence at only 18 of the 25 km² grid squares on the South African distribution map since 2015. Present in all seven survey seasons at Swadini, where it favours shaded riverbank localities with overhanging vegetation reaching the water. The Swadini population is one of the best-documented sites for this species in South Africa.



Phaon iridipennis — Glistening Demoiselle (female) · Photo: A. de Vries

***Aciagrion gracile* — Graceful Slim**

Coenagrionidae · Zygoptera · 37–39 mm

A forest-floor specialist observed in deep shade beneath the canyon's indigenous canopy. Found on moist ground or fluttering slowly through mottled sunlight, foraging for small insects. Winter individuals display striking **salmon-pink colouration** that gradually transitions to the adult blue phase by late August — an unusually slow maturation process documented for the first time at this site. By mid-September 2021, returning visitors noted substantial blue colouration already developing on the thorax and abdomen.

Crocothemis divisa — Rock Scarlet	Orthetrum brachiale — Banded Skimmer
Libellulidae · Anisoptera · 36–39 mm	Libellulidae · Anisoptera · 43–47 mm
Seldom recorded; identified by diagnostic wing venation and complete absence of amber at hindwing bases. A single young male photographed in 2019 (week 25) on riverside rocks. Likely more frequent during the unsurveyed January–March summer peak.	A little-known species rarely encountered across its range. Recorded once at Swadini in 2022 (week 21), confirmed by two observers with photographic evidence. Its presence underscores the value of sustained multi-year surveys at a single site.

7. Complete Species Checklist

All 70 identified species recorded at Swadini across seven seasons (2019–2025). ● = recorded – = not recorded.
★ = nationally notable. Yrs column: dark green = 7 seasons; medium green = 5–6; light green = 3–4; pink = 1–2.



Platycypha caligata



Notiothemis jonesi



Tetrathemis pollenii · Photo: Joey de Vries



Crocothemis sanguinolenta

Scientific Name	Common Name	19	20	21	22	23	24	25	Yrs
Aeshnidae · Emperors & Hawkers									
<i>Anax imperator</i>	Blue Emperor	●	●	●	–	●	–	●	5
<i>Anax speratus</i>	Orange Emperor	●	●	●	●	●	●	●	7
<i>Pinheyschna subpupillata</i>	Stream Hawker	●	●	●	●	–	–	–	4
Gomphidae · Clubtails									
<i>Ceratogomphus pictus</i>	Common Thorntail	●	–	–	–	–	–	–	1
<i>Ictinogomphus ferox</i>	Common Tigertail	●	●	–	–	–	–	–	2
<i>Lestinigomphus angustus</i>	Spined Fairytail	●	–	●	●	●	●	●	6
<i>Microgomphus nyassicus</i> ★	Eastern Scissortail	●	–	●	●	●	●	●	6
<i>Onychogomphus supinus</i>	Lined Claspertail	–	–	●	●	●	–	●	4
<i>Paragomphus cognatus</i>	Rock Hooktail	●	–	–	●	●	●	–	4
<i>Paragomphus genei</i>	Common Hooktail	–	–	●	–	●	–	–	2
Macromiidae · Cruisers									
<i>Phyllomacromia picta</i>	Darting Cruiser	●	●	●	●	●	●	●	7
Libellulidae · Skimmers, Dropwings & allies									
<i>Acisoma inflatum</i>	Stout Pintail	●	●	–	–	–	–	–	2
<i>Acisoma variegatum</i>	Slender Pintail	–	●	–	–	–	–	–	1
<i>Brachythemis lacustris</i>	Red Groundling	●	●	●	–	●	–	–	4
<i>Brachythemis leucosticta</i>	S. Banded Groundling	●	–	–	–	●	●	●	4
<i>Bradinopyga cornuta</i>	Horned Rockdweller	●	●	–	–	–	–	–	2
<i>Crocothemis divisa</i> ★	Rock Scarlet	●	–	–	–	–	–	–	1
<i>Crocothemis erythraea</i>	Broad Scarlet	●	●	●	●	●	●	●	7
<i>Crocothemis sanguinolenta</i>	Little Scarlet	●	●	●	●	●	●	●	7
<i>Diplacodes lefebvreii</i>	Black Percher	–	–	●	–	–	–	–	1
<i>Hemistigma albipunctum</i>	African Piedspot	●	●	–	●	–	–	–	3
<i>Nesciothemis farinosa</i>	Eastern Blacktail	●	●	●	●	●	●	●	7
<i>Notiothemis jonesi</i>	Eastern Forestwatcher	●	●	●	●	–	●	–	5
<i>Orthetrum abbotti</i>	Little Skimmer	●	●	●	–	●	–	–	4
<i>Orthetrum brachiale</i> ★	Banded Skimmer	–	–	–	●	–	–	–	1
<i>Orthetrum caffrum</i>	Two-striped Skimmer	–	–	–	●	–	–	–	1
<i>Orthetrum chrysostigma</i>	Epaulet Skimmer	●	●	●	●	●	●	●	7

Scientific Name	Common Name	19	20	21	22	23	24	25	Yrs
<i>Orthetrum hintzi</i>	Dark-shouldered Skimmer	●	—	—	—	—	—	—	1
<i>Orthetrum julia</i>	Julia Skimmer	●	●	●	●	●	●	●	7
<i>Orthetrum machadoi</i>	Highland Skimmer	●	—	—	—	—	—	—	1
<i>Orthetrum stemmale</i>	Bold Skimmer	—	●	●	●	●	●	—	5
<i>Orthetrum trinacria</i>	Long Skimmer	—	—	—	●	—	—	—	1
<i>Palpopleura jucunda</i>	Yellow-veined Widow	●	●	●	—	—	●	—	4
<i>Palpopleura lucia</i>	Lucia Widow	●	●	●	●	●	●	●	7
<i>Palpopleura portia</i>	Portia Widow	●	●	●	●	●	●	—	6
<i>Pantala flavescens</i>	Pantala	—	—	—	—	—	—	●	1
<i>Tetrathemis pollenii</i>	Black-splashed Elf	●	●	●	●	●	●	●	7
<i>Tramea basilaris</i>	Keyhole Glider	—	—	—	●	—	—	—	1
<i>Trithemis aconita</i>	Halfshade Dropwing	●	●	●	●	●	●	●	7
<i>Trithemis annulata</i>	Violet Dropwing	●	—	—	—	—	—	—	1
<i>Trithemis arteriosa</i>	Red-veined Dropwing	●	●	●	●	●	●	●	7
<i>Trithemis donaldsoni</i>	Denim Dropwing	●	●	●	●	●	●	●	7
<i>Trithemis furva</i>	Navy Dropwing	—	—	●	—	●	—	—	2
<i>Trithemis kirbyi</i>	Orange-winged Dropwing	●	●	●	●	●	●	●	7
<i>Trithemis pluvialis</i>	Russet Dropwing	●	●	●	●	●	●	●	7
<i>Trithemis stictica</i>	Jaunty Dropwing	●	●	—	●	—	—	—	3
<i>Zygonoides fuelleborni</i>	Southern Riverking	●	—	—	—	—	—	—	1
<i>Zygonyx natalensis</i>	Blue Cascader	●	●	●	●	—	—	●	5
<i>Zygonyx torridus</i>	Ringed Cascader	●	—	●	—	—	—	—	2
Calopterygidae · Demoiselles									
<i>Phaon iridipennis</i>	Glistening Demoiselle	●	●	●	●	●	●	●	7
Chlorocyphidae · Jewels									
<i>Platycypha caligata</i>	Dancing Jewel	●	●	●	●	●	●	●	7
Lestidae · Spreadwings									
<i>Lestes pallidus</i>	Pallid Spreadwing	—	—	●	—	—	—	—	1
<i>Lestes plagiatus</i>	Highland Spreadwing	●	●	●	●	●	●	●	7
<i>Lestes virgatus</i>	Smoky Spreadwing	—	●	●	●	●	●	●	6
Coenagrionidae · Sprites, Slims & allies									
<i>Aciagrion gracile</i> ★	Graceful Slim	—	—	●	—	●	●	—	3
<i>Azuragrion nigradorsum</i>	Sailing Bluet	—	—	●	—	—	—	—	1
<i>Ceriagrion glabrum</i>	Common Citril	●	—	●	—	●	●	—	4
<i>Ischnura senegalensis</i>	Tropical Bluetail	—	—	●	—	—	—	●	2
<i>Pseudagrion commoniae</i>	Black Sprite	●	●	●	●	—	—	—	4
<i>Pseudagrion gamblesi</i> ★	Great Sprite	●	●	●	●	●	●	●	7
<i>Pseudagrion hageni</i>	Painted Sprite	●	●	●	●	●	●	—	6
<i>Pseudagrion hamoni</i>	Swarthy Sprite	●	●	●	●	●	●	●	7
<i>Pseudagrion kersteni</i>	Powder-faced Sprite	●	●	●	●	●	●	●	7
<i>Pseudagrion massaicum</i>	Masai Sprite	●	—	●	—	—	—	—	2
<i>Pseudagrion salisburyense</i>	Slate Sprite	—	—	●	●	●	—	—	3
<i>Pseudagrion sjoestedti</i>	Variable Sprite	●	●	●	●	●	●	●	7

Scientific Name	Common Name	19	20	21	22	23	24	25	Yrs
<i>Pseudagrion sublacteum</i>	Cherry-eye Sprite	●	●	●	●	●	●	●	7
<i>Pseudagrion sudanicum</i>	Blue-sided Sprite	–	●	●	●	●	–	–	4
Platycnemididae · Threadtails & Riverjacks									
<i>Elatoneura glauca</i>	Common Threadtail	●	●	●	●	●	●	●	7
<i>Mesocnemis singularis</i>	Savanna Riverjack	●	●	●	●	●	●	●	7

● = recorded | – = not recorded | ★ = nationally notable species | Common names per Wilkinson & Dickinson (dragonflies-id.co.za)

8. Conservation Value & Recommendations

The Odonata survey data collectively make a compelling case for the outstanding conservation value of the Blyde River Canyon at Swadini:

Exceptional diversity for a small area. 70 species within 1.1 km² represents ~43% of the national checklist in a single canyon site — an internationally significant figure for a freshwater invertebrate group.

First South African living adult of *Microgomphus nyassicus*. The confirmation of this species as a breeding resident of the Blyde River Canyon, with copulating pairs on record from 2021 and 2025, is a discovery of national scientific significance that warrants formal publication.

Year-round biodiversity asset. 20–33 species remain on the wing through the winter months (June–August), making the canyon viable for Odonata observation during any visit throughout the survey period.

Drought sensitivity signal. The sharp drop in 2024 across all families provides a measurable baseline against which future rainfall events can be correlated. Integration of the long-term rainfall dataset (J. Pienaar, from 2000) will strengthen this analysis considerably.

Survey gaps worth addressing. The January–March summer peak is entirely unrepresented. Even a single season of comprehensive summer coverage would likely add 20–30+ species to the site list and document the true annual diversity maximum.

Photography and citizen science. All records are photographic and lodged with two accredited databases. This dataset is citable, reproducible, and ready for integration into formal biodiversity assessments of the Blyde River Canyon Nature Reserve.

Photography & Data Ownership

All photographs in this report were taken by **Andries de Vries** at Swadini, Blyde River Canyon, with the exception of *Tetrathemis pollenii* (photographed by Joey de Vries). Photographs are the intellectual property of the respective photographers.

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Report generated: April 2026 · Survey data: May 2019 – 27 August 2025 · Total records: 3,396 · Species: 70 · Families: 9